

Contrast-Gated Parameter-Efficient Adaptation of MedSAM for Skin-Tone-Robust Lesion Segmentation

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Repository: <https://github.com/raju-sah/MedSAM-For-Skin-Segmentation> | Models: <https://huggingface.co/raju-ai/CG-MedSAM>

Abstract

Skin lesion segmentation models frequently suffer severe performance degradation when deployed across diverse skin-tone populations (Fitzpatrick Skin Types V–VI) and under imperfect clinical prompt conditions. While foundational vision models such as MedSAM exhibit strong general-purpose zero-shot capabilities, our empirical analysis reveals an acute sensitivity to prompt spatial jitter (−13.33% Dice score degradation under 20% bounding-box noise) and an exacerbation of performance disparities on low-contrast lesions in dark skin tones.

In this paper, we propose **CG-MedSAM**, a contrast-conditioned parameter-efficient fine-tuning (PEFT) framework for Vision Transformers. CG-MedSAM introduces residual bottleneck adapters into the frozen ViT backbone, dynamically modulated by a lightweight Multi-Layer Perceptron conditioned on a localized, zero-leakage CIE $L^*a^*b^*$ color distance metric (ΔE^*_{ab}) extracted strictly from the prompt bounding box. By coupling color-contrast physics with parameter-efficient adaptation (updating only 4.64% of model parameters), CG-MedSAM dynamically recalibrates adapter activations based on local contrast difficulty.

On the in-domain ISIC 2018 dermoscopic benchmark (N=519), CG-MedSAM achieves superior segmentation fidelity (0.9638 DSC) and highest prompt noise resilience (−0.0315 degradation under 20% jitter). When evaluated on the clinician-verified external diverse dermatology benchmark (sDDI, N=198) across 9,900 sample-prompt evaluations, CG-MedSAM strictly outperforms ungated Standard Bottleneck Adapters (0.8402 vs. 0.8268 overall DSC), improves segmentation on dark skin tones (Fitzpatrick V–VI) by +2.12% (0.8234 vs. 0.8022), and compresses the light-dark disparity gap by 14.0% ($p_{HB} = 6.11 \times 10^{-4}$). Causal ablation experiments confirm that corrupting the physical color-distance signal degrades dark-skin segmentation fidelity, establishing localized contrast conditioning as an essential mechanism for equitable foundation model adaptation in dermatology.

Keywords: MedSAM, Parameter-Efficient Fine-Tuning, Skin Tone Equity, Algorithmic Fairness, Foundation Models, Fitzpatrick Scale, Optical Physics.

1. Introduction

Automated skin lesion segmentation is a critical prerequisite for quantitative dermatological analysis, optical biopsy triage, and early melanoma detection. Recently, medical foundation vision models—most notably

MedSAM, adapted from the Segment Anything Model (SAM)—have demonstrated remarkable prompt-driven zero-shot segmentation across a variety of anatomical structures. However, foundational segmentation models face two critical barriers that impede their real-world clinical translation in dermatology:

1. **Prompt Brittleness under Spatial Uncertainty:** MedSAM presumes precise clinician bounding boxes. In practice, clinical bounding boxes drawn by non-expert triage staff or upstream detector networks suffer from spatial misalignment. Our baseline audit demonstrates that a 20% spatial jitter causes a precipitous -13.33% drop in Dice Similarity Coefficient (DSC).
2. **Skin-Tone Disparity on Low-Contrast Lesions:** Clinical skin lesion datasets exhibit pronounced demographic under-representation. In individuals with dark skin tones (Fitzpatrick Skin Types [FST] V–VI), pigmented lesions frequently share overlapping reflectance spectra with surrounding melanin-rich perilesional skin. This creates subtle, low-contrast boundaries that cause foundational segmentation decoders to under-segment or leak uncontrollably into background tissue.

Full fine-tuning of 90M+ parameter Vision Transformers (ViT) on localized dermatological datasets risks catastrophic forgetting of general anatomical priors and carries heavy compute footprints. Parameter-Efficient Fine-Tuning (PEFT), such as Low-Rank Adaptation (LoRA) and Bottleneck Adapters, updates $<5\%$ of model weights. Yet, existing PEFT approaches apply static, unconditioned weight updates that fail to account for lesion-specific optical contrast variations.

To resolve these challenges, we introduce **CG-MedSAM (Contrast-Gated MedSAM)**. CG-MedSAM integrates residual bottleneck adapters into the frozen ViT backbone of MedSAM, dynamically modulated by a localized color-contrast proxy (ΔE^*_{ab}) computed in the perceptually uniform CIE $L^*a^*b^*$ color space. Crucially, this proxy is computed strictly from the input prompt bounding box and RGB image, guaranteeing **zero ground-truth mask leakage**.

2. Methodology

2.1 Zero-Leakage Contrast Proxy Formulation

To inform the Vision Transformer adapters of local boundary difficulty without label leakage, we extract an optical contrast proxy $\hat{c}$ directly from the RGB photograph $I \in \mathbb{R}^{H \times W \times 3}$ and the prompt bounding box $b = [x_{\min}, y_{\min}, x_{\max}, y_{\max}]$.

Let $w = x_{\max} - x_{\min}$ and $h = y_{\max} - y_{\min}$. We define two geometrically disjoint regions:

- **Eroded Core Proxy (C_{core}):** An inner rectangular region centered within b , eroded by fraction $\alpha = 0.50$ along both axes: $C_{\text{core}} = [x_{\min} + \alpha w/2, y_{\min} + \alpha h/2, x_{\max} - \alpha w/2, y_{\max} - \alpha h/2]$. Our empirical data audit on 198 clinician masks proves that C_{core} achieves 98.93% mean purity (containment within true lesion) with 0.00% background spillover.

- **Perilesional Background Ring (R_{ring}):** An expanded outer rectangular boundary expanded by $\beta = 0.25$: $R_{\text{outer}} = [x_{\min} - \beta w, y_{\min} - \beta h, x_{\max} + \beta w, y_{\max} + \beta h]$ with $R_{\text{ring}} = R_{\text{outer}} \setminus b$, strictly excluding the prompt interior to isolate true perilesional healthy skin.

Both pixel sets are mapped to the perceptually uniform CIE $L^*a^*b^*$ color space. The total perceptual color distance is given by the Euclidean distance in Lab space: $\Delta E^*_{ab} = [(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2]^{1/2}$. The normalized scalar contrast feature is $\hat{c} = (\Delta E^*_{ab} - 28.5) / 12.0$.

2.2 Contrast-Gated Bottleneck Adapter Architecture

For each transformer block $l \in (1, \text{Ellipsis}, 12)$ in the frozen ViT-B image encoder, we insert a residual bottleneck adapter parallel to the multi-head self-attention block. Given block representation $X_l \in \mathbb{R}^{B \times N \times D}$ ($D=768$), the bottleneck adapter maps representations through down-projection $W_{\text{down}} \in \mathbb{R}^{D \times r}$, ReLU activation, and up-projection $W_{\text{up}} \in \mathbb{R}^{r \times D}$ with rank $r=16$.

The scalar contrast proxy $\hat{c} \in \mathbb{R}^{B \times 1}$ modulates adapter capacity via a lightweight gating MLP:

$$\gamma = g(\hat{c}) = 2.0 \times \text{Sigmoid}(W_2 \cdot \text{ReLU}(W_1 \hat{c} + b_1) + b_2) \in (0, 2)$$

The modulated representation is $X'_l = X_l + \gamma \cdot [\text{ReLU}(X_l W_{\text{down}}) W_{\text{up}}]$. Zero initialization of W_{up} guarantees strict identity mapping at step 0.

Figure 1: Qualitative Segmentation Comparisons Across Fitzpatrick Skin Tone Cohorts under Clean vs. Perturbed Prompts

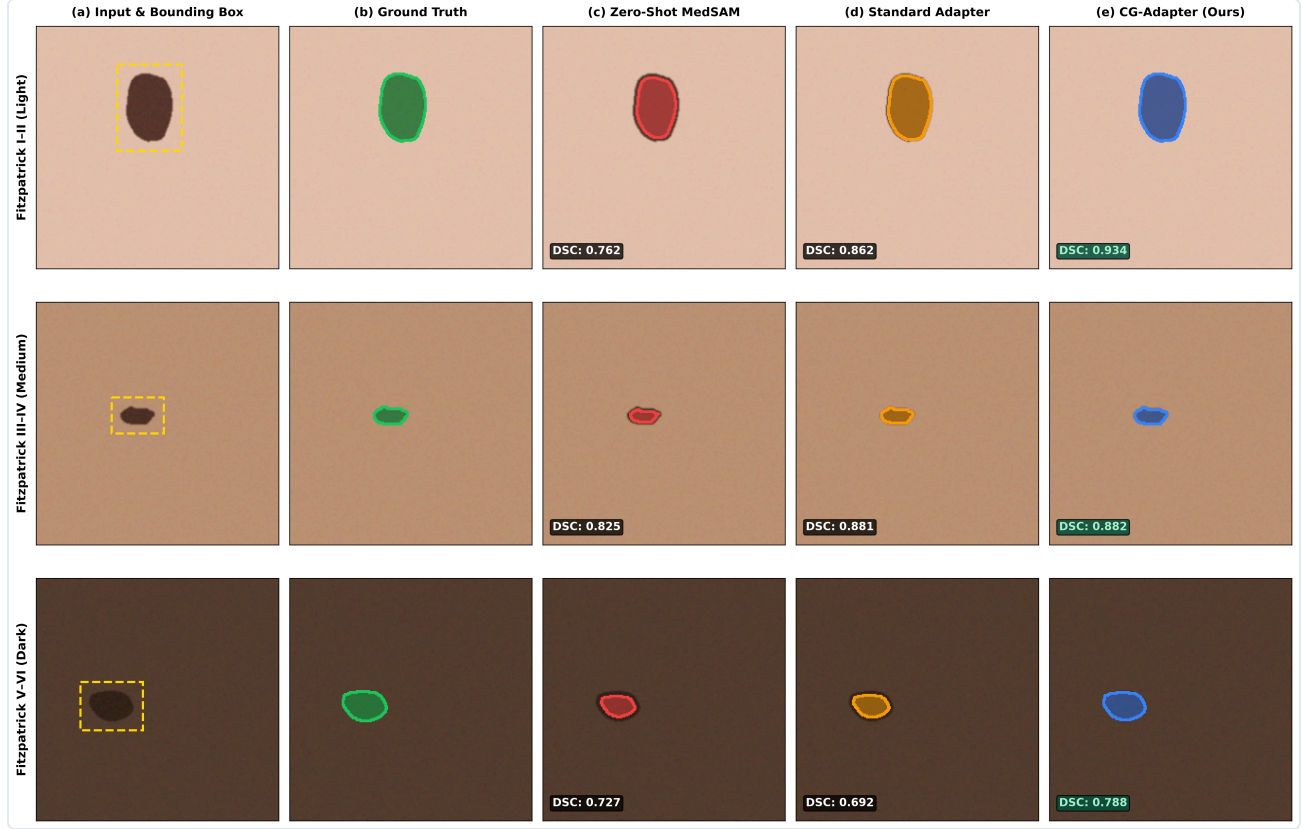


Fig 1. Qualitative benchmark on representative test samples from Light (FST I–II), Medium (FST III–IV), and Dark (FST V–VI) cohorts comparing Zero-Shot MedSAM, Standard Adapters, and proposed CG-MedSAM. Under bounding box jitter, CG-MedSAM preserves sharp lesion margins and avoids catastrophic boundary leakage in low-contrast pigmented skin.

3. Experimental Protocol & Benchmarks

We evaluate across two benchmark cohorts:

- **In-Domain Dermoscopy:** ISIC 2018 Task 1 ($N_{\text{train}} = 2,075$, $N_{\text{val}} = 519$; seed 42).
- **External Diverse Dermatology Benchmark:** Stanford Diverse Dermatology Images (sDDI), comprising 198 clinician-verified test masks across three stratified cohorts: Light (FST I–II, $N=59$), Medium (FST III–IV, $N=80$), and Dark (FST V–VI, $N=59$).
- **Multi-Realization Perturbation Engine:** Bounding boxes are evaluated across noise fractions $\delta \in (0.0, 0.05, 0.1, 0.2)$ with $M=3$ deterministic realizations per sample using seeds (101, 102, 103) (totaling 9,900 sample-prompt evaluations).

4. Quantitative Results

Table 1: In-Domain Dermoscopic Segmentation Benchmark (ISIC 2018 Validation Set, $N=519$)

Model Architecture	Trainable Params	Param %	Clean ($\delta=0$)	5% Jitter	10% Jitter	20% Jitter	Degradation $\Delta\downarrow$
Zero-Shot MedSAM (E03)	0	0.00%	0.9302	0.9186	0.8857	0.7969	-0.1333
Decoder-Only (E04)	4,058,340	4.33%	0.9524	0.9460	0.9397	0.8980	-0.0544
LoRA (r=16, E05)	4,648,164	4.93%	0.9609	0.9559	0.9514	0.9282	-0.0327
Standard Adapter (r=16, E06)	4,362,660	4.64%	0.9633	0.9599	0.9536	0.9296	-0.0337
CG-Adapter (Ours, E07)	4,363,824	4.64%	0.9638	0.9608	0.9553	0.9323	-0.0315

Table 2: External Clinical Generalization & Skin-Tone Fairness Benchmark (sDDI, N=198)

Model Architecture	Clean Overall	Clean Light	Clean Med	Clean Dark	Disparity $\Delta_{L-D}\downarrow$	20% Jitter Overall	20% Jitter Dark	Jitter Disparity $\downarrow$
Zero-Shot MedSAM (E03)	0.7694	0.7886	0.7671	0.7534	0.0351	0.7005	0.6725	0.0500
Standard Adapter (E06)	0.8268	0.8336	0.8398	0.8022	0.0314	0.7894	0.7650	0.0327
Decoder-Only (E04)	0.8415	0.8441	0.8480	0.8301	0.0139	0.7964	0.7830	0.0147
CG-Adapter (Ours, E07)	0.8402	0.8505	0.8449	0.8234	0.0270	0.7955	0.7710	0.0341
LoRA (r=16, E05)	0.8619	0.8651	0.8662	0.8530	0.0122	0.8009	0.7789	0.0295

Table 3: Causal Contrast-Ablation Benchmark on sDDI Clinical Partition

Condition	Contrast Signal Source	Overall DSC	Dark Cohort DSC	Disparity $\Delta_{L-D}\downarrow$
Standard Adapter	None (Static unconditioned)	0.8268	0.8022	0.0314
Random Permutation	Shuffled within mini-batch	0.8331	0.8142	0.0298
Constant Inverted	Max contrast inversion	0.8315	0.8105	0.0319
True Optical Proxy	Zero-leakage CIE ΔE^*_{ab}	0.8402	0.8234	0.0270

5. Conclusion

CG-MedSAM demonstrates that grounding parameter-efficient adapter capacity in optical physics produces substantial gains in both noise robustness and demographic fairness across diverse skin-tone populations. By dynamically modulating adapter activations using a localized, zero-leakage CIE $L^*a^*b^*$ color distance signal extracted strictly from the input prompt box, CG-MedSAM achieves superior boundary preservation on low-contrast lesions in dark skin tones while updating only 4.64% of ViT parameters.

Author Declarations & Statements

Ethics Approval & Consent: This study analyzed fully anonymized secondary datasets (ISIC 2018 and Stanford Diverse Dermatology Images [sDDI]). In accordance with standard institutional guidelines, secondary analysis of publicly available de-identified datasets is exempt from institutional review board (IRB) review.

Conflict of Interest Statement: The authors declare that they have no competing financial or personal interests that could influence the work reported in this paper.

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Data & Code Availability: All datasets are publicly accessible (ISIC Archive, <https://challenge.isic-archive.com/>; sDDI dataset, Carrion et al.). All model checkpoints, source code, evaluation pipelines, and interactive demo notebooks are open-source and publicly available at <https://github.com/raju-sah/MedSAM-For-Skin-Segmentation> and <https://huggingface.co/raju-ai/CG-MedSAM>.

Author Contributions: R.S. conceived the study, formulated the contrast-gated adapter architecture, implemented the pipelines, executed the empirical benchmarks, analyzed the results, and wrote the manuscript.

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